

Project Title

Resume–Job Matching & Talent Marketplace Engine

Project Report submitted for the partial fulfillment of grade for the course

Data Structures and Algorithms(25CS2103E)

in

A.Y. 2026-2027

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ABSTRACT

The Résumé–Job Matching & Talent Marketplace Engine is a comprehensive platform designed to simulate real-world recruitment systems by integrating multiple functional modules. In today’s dynamic job market, students often lack clarity on their eligibility, while recruiters face challenges in screening numerous résumés. The system includes a student module for profile creation and résumé upload, where key information such as skills, education, and projects is extracted, and a recruiter module for posting job requirements.

The core component is the matching engine, which uses data structures and algorithms like hashing, string matching, and priority queues to compare skills and rank candidates based on compatibility scores. Additional features such as skill gap analysis, career roadmap generation, and mock interview preparation enhance its practical value, helping students improve employability while enabling efficient and accurate candidate selection for recruiters.

Signature of the Guide